

CMSC 207: Chapter 8 Homework

May 3, 2026

1. Let R be a relation defined on the set of all integers $\mathbb{Z}$ as follows:

$$\text{for all integers } m \text{ and } n, m R n \iff 5 \mid (m - n).$$

Prove that R is an equivalence relation on $\mathbb{Z}$.

2. Let S be the set of all strings of 0s and 1s of length 3. Define a relation R on S as follows:

$$\text{for all strings } s \text{ and } t \in S, s R t \iff \text{the two left-most characters of } s \text{ and } t \text{ are the same.}$$

Prove that R is an equivalence relation on S .